collatz_bit_tracker

Massless Information-Entropy Collatz Framework • Python Package Technical Reference

PACKAGE VERSION	AUTHOR	LICENSE	PYTHON REQUIREMENT
1.0.0	Cody Crawford	MIT License	≥ 3.6

1. Package Overview

collatz_bit_tracker is a specialized computational Python library engineered for low-level bitwise tracking within the *Massless Information-Entropy Collatz Framework*. The library provides optimized data structures and visual tools to trace bit shifts, trajectories, and binary state entropy evolution during iterative Collatz sequence generation.

2. Standard Distribution Setup (`setup.py`)

```
from setuptools import setup, find_packages

setup(
    name="collatz_bit_tracker",
    version="1.0.0",
    author="Cody Crawford",
    description="A computational bitwise tracking library for the Massless Information-
Entropy Collatz Framework.",
    long_description=open("README.md").read(),
    long_description_content_type="text/markdown",
    url="https://github.com",
    packages=find_packages(),
    classifiers=[
        "Programming Language :: Python :: 3",
        "License :: OSI Approved :: MIT License",
        "Operating System :: OS Independent",
        "Topic :: Scientific/Engineering :: Mathematics",
    ],
    python_requires=">=3.6",
    install_requires=[
        "matplotlib>=3.0.0",
    ],
)
```

3. Metadata & Dependencies Specifications

Parameter	Configuration	Description
Package Name	`collatz_bit_tracker`	Registered Python package identifier
Core Dependency	`matplotlib ≥ 3.0.0`	Required for bit distribution & entropy plots